

MULTI-STEP PROMPT WORKFLOW

Mini Project: Smart Canteen Management System

Name: Nallapureddy Mohan Reddy

Registration Number: 99230040690

Scenario: Transforming a mini-project idea into a structured project proposal

1. Project Overview

Project Title: Smart Canteen Management System

Smart Canteen Management System is a simple web-based application that helps students and canteen staff manage food items, place orders, record order status, and maintain basic customer and menu information. The project aims to reduce manual order recording and provide a clear view of daily canteen transactions.

Suggested Technology: HTML, CSS, JavaScript, Java, MySQL

Users: Students, canteen staff, and administrators

Core Features: Food item entry, menu management, item search, order placement, order status records, database storage, order history, and basic sales reports.

2. Raw Project Notes

College canteens often record food orders and menu details manually or through separate records. During busy hours, it becomes difficult to track available items, pending orders, completed orders, and daily sales accurately. The proposed Smart Canteen Management System should allow staff to add food items, manage the menu, allow students to place orders, record order status, save information in a database, and view simple reports. The interface should be easy for beginners to use.

STEP 1 — ANALYSIS & PROJECT PLANNING

Exact Prompt Used

Act as an expert software project analyst and technical documentation writer.

I want to develop a mini project called "Smart Canteen Management System".

Analyze the raw project notes below and produce:

1. The main problem statement.
2. Exactly 4 key project objectives.
3. The target users.
4. Exactly 6 core functional requirements.
5. A simple technology stack using HTML, CSS, JavaScript, Java, and MySQL.
6. A structured outline for a 700-word mini-project proposal.

Do not invent advanced features that are not supported by the notes.

Output format:

- Problem Statement
- Objectives
- Target Users
- Functional Requirements
- Technology Stack
- Proposal Outline

Raw notes:

College canteens often record food orders and menu details manually or through separate records. It becomes difficult to track available items, pending orders, completed orders, and daily sales accurately. The proposed Smart Canteen Management System should allow staff to add food items, manage the menu, allow students to place orders, record order status, save information in a database, and view simple reports. The interface should be easy for beginners to use.

Intermediate AI Response — Live Execution Record

- Problem Statement: Manual canteen order and menu records make it difficult to track food availability, pending/completed orders, and daily sales accurately.

- Objectives:

1. Digitize food item and menu management.
2. Simplify student food-order placement.
3. Maintain accurate order and status records.
4. Store transaction information and provide basic reports.

- Target Users: Students, canteen staff, and administrators.

- Functional Requirements:

1. Add and manage food items.
2. Maintain and display the menu.

3. Search available food items.
 4. Place and record student orders.
 5. Update and view order status.
 6. Store order data and generate basic reports.
- Technology Stack: HTML, CSS, JavaScript, Java, MySQL.
 - Proposal Outline: Introduction → Problem Statement → Objectives → Proposed Solution → Key Features → Technology Stack → Expected Benefits → Conclusion.

STEP 2 — PROPOSAL GENERATION

Exact Prompt Used

Excellent. Using only the project analysis, objectives, requirements, technology stack, and proposal outline from your previous response, write a complete mini-project proposal for the Smart Canteen Management System.

Strict constraints:

1. Maximum 700 words.
2. Use a professional academic tone.
3. Include: Introduction, Problem Statement, Objectives, Proposed Solution, Key Features, Technology Stack, Expected Benefits, and Conclusion.
4. Explain how the system works in simple terms.
5. Do not introduce features or technologies that were not included in Step 1.
6. Use short paragraphs of no more than 3 sentences each.
7. Include this simple text-based workflow:
User → Web Interface → Java Application → MySQL Database → Canteen Report.
8. Make the proposal suitable for a college mini-project submission.

Output only the first draft of the proposal.

Intermediate AI Response — Live Execution Record

Introduction: The Smart Canteen Management System is a simple web-based application designed to organize food menu and order activities in a college canteen. It provides students and staff with a structured way to manage food items, place orders, track order status, and maintain transaction information. **Problem Statement:** Many small college canteens depend on manual records for food items and orders. This can make it difficult to identify available items, track pending and completed orders, maintain order history, and prepare daily reports accurately.

Objectives: The project aims to digitize menu management, simplify food-order placement, maintain accurate order status records, and store transaction information for basic reporting. **Proposed Solution:** The system provides a simple interface through which canteen staff can add and manage food items and students can view the menu, search items, and place orders. Staff can record and update order status, while the system stores the information in MySQL for later retrieval and reporting. **Key Features:** The main features are food item entry, menu management, food-item search, order placement, order status recording, database storage, order history, and basic canteen reports. These functions are focused on the requirements of a small college canteen. **Technology Stack:** HTML will structure the web pages, CSS will provide presentation and layout, and JavaScript will support basic interaction. Java will implement the application logic, while MySQL will store food, menu, and order information. **Workflow:** User → Web Interface → Java Application → MySQL Database → Canteen Report. **Expected Benefits:** The system can reduce manual order recording, make menu information easier to manage, and provide a clearer view of order status and stored transactions. It can also help staff prepare simple reports from the recorded information. **Conclusion:** The Smart Canteen Management System provides a practical mini-project approach to organizing canteen menu and order activities. By using HTML, CSS, JavaScript, Java, and MySQL, the project demonstrates how a simple web-based application can improve basic record management.

STEP 3 — CRITIQUE & REFINEMENT

Exact Prompt Used

Act as a strict senior project reviewer. Critically evaluate the proposal you just wrote against the following checklist:

1. Is it 700 words or fewer?
2. Does it contain all required sections?
3. Are exactly 4 objectives represented?
4. Are the six core functional requirements from Step 1 accurately reflected?
5. Does it use only the approved technology stack: HTML, CSS, JavaScript, Java, and MySQL?
6. Is the workflow clear and technically consistent?
7. Are unsupported or exaggerated claims avoided?
8. Is the language suitable for an academic mini-project submission?
9. Are paragraphs limited to a maximum of 3 sentences?
10. Is the conclusion clear?

First, provide a "Self-Critique" with a bullet-by-bullet evaluation of all 10 criteria. Identify any weaknesses and explain how they should be fixed.

Then provide "Final Polished Mini-Project Proposal". Correct every issue you identify while preserving the useful content from the previous draft.

The final proposal must remain within 700 words and satisfy all 10 criteria. Do not skip the self-critique.

Intermediate AI Response — Live Execution Record

Final Polished Mini-Project Proposal

Introduction: The Smart Canteen Management System is a simple web-based application for organizing food menu and order activities in a college canteen. It helps students and staff manage food information, orders, order status, and stored transaction records. **Problem Statement:** Manual canteen records can make it difficult to track available food items, pending orders, completed orders, and daily transactions. A simple computerized system is therefore needed to organize these records and make them easier to retrieve. **Objectives:** The project has four objectives: digitize food-item and menu management; simplify student order placement; maintain accurate order-status records; and store transaction information for basic reports. **Proposed Solution:** Canteen staff can add food items and maintain the menu, while students can view and search available items and place orders. The system records each order, allows staff to update its status, and stores the information in a MySQL database. **Key Features:** The six core functional requirements are food-item management, menu management, food-item search, order placement, order-status management, and database storage with basic reports. Together, these features cover the main activities described in the project requirements. **Technology Stack:** HTML is used to create the web structure, CSS to design the interface, and JavaScript for basic interaction. Java handles application logic, while MySQL stores food, menu, and order information. **Workflow:** User → Web Interface → Java

Application → MySQL Database → Canteen Report. **Expected Benefits:** The system can reduce manual order recording and make food and order information easier to manage. It also provides a clearer record of order status and stored transactions for basic reporting. **Conclusion:** The Smart Canteen Management System is a practical college mini-project for applying web, Java, and database concepts to a real-world canteen scenario. Its focused features provide a simple approach to managing menu and order records without introducing unnecessary advanced technologies.

4. Chaining Dependency

Step 1 → Step 2 → Step 3. Step 1 transforms the raw canteen project notes into a structured analysis. Step 2 explicitly uses the Step 1 output to generate a complete proposal. Step 3 evaluates the Step 2 proposal against a defined checklist and transforms it into a corrected final version.

5. Critique Gate

Step 3 is the required critique gate. The AI first reviews its previous proposal against ten specific criteria, identifies weaknesses, and then produces a refined final version that addresses the identified issues.

6. Live Execution Evidence

Run all three prompts sequentially in the same assigned AI interface. Add screenshots showing Prompt 1 and its response, Prompt 2 and its response, and Prompt 3 with the self-critique and final response. If the platform provides a shareable chat link, include it in the submission.

7. Assessment Checklist

Requirement	Evidence in this workflow
3+ chained prompts	Three sequential prompts are provided.
Explicit dependency	Steps 2 and 3 explicitly use the preceding outputs.
Critique gate	Step 3 contains a 10-point review and correction stage.
Mini-project scenario	Smart Canteen Management System is used as the project.
Live testing	Actual outputs and screenshots should be inserted after running the prompts.

8. Submission Note

Replace each 'Live Execution Record' placeholder with the exact AI-generated output obtained by running the three prompts in sequence. Add screenshots or the chat link as required by the assessment instructions.